

ReLU and Related Activation Functions

Task 1 | Formulas, gradient behavior and practical tradeoffs

Nonlinear activations let stacked neural layers represent nonlinear relationships. ReLU is popular because it uses a simple comparison, passes positive inputs without saturation, and creates sparse outputs. Its zero negative-side derivative can leave a unit inactive on all relevant inputs. The following variants change gradient flow, regularization or output range; no variant is universally best.

Definitions and input behavior

Define $r_a(x) = x$ for $x > 0$ and ax for $x \leq 0$, and $e_\alpha(x) = x$ for $x > 0$ and $\alpha(\exp(x)-1)$ otherwise. The following equations cover all required functions. Values at zero are zero; derivatives at kinks use implementation conventions.

Function	Mathematical definition	Positive and negative behavior
ReLU	$f(x) = \max(0, x)$	Positive: identity. Negative: exactly zero.
Leaky ReLU	$f(x) = r_a(x)$, fixed $a > 0$	Positive: identity. Negative: small linear slope, commonly $a = 0.01$.
PReLU	$f(x) = r_a(x)$, learned a	Positive: identity. Negative: slope learned per channel or shared.
RReLU	$f(x) = r_a(x)$ Training: $a \sim U(l, u)$ Evaluation: $a = (l + u)/2$	Positive: identity. Negative: stochastic slope in training, deterministic mean slope in evaluation.
ELU	$f(x) = e_\alpha(x)$, $\alpha > 0$	Positive: identity. Negative: smooth exponential branch tending to $-\alpha$.
SELU	$f(x) = \lambda e_\alpha(x)$ $\alpha \approx 1.67326$, $\lambda \approx 1.05070$	Positive: λx . Negative: tends to $-\lambda\alpha$. Constants target self-normalization under assumptions.
ReLU6	$f(x) = \min(\max(0, x), 6)$	Zero below 0; identity between 0 and 6; capped at 6 above 6.
GELU	$f(x) = x\Phi(x)$ Φ : standard-normal CDF	Positive: smoothly approaches identity. Negative: small negative outputs, tending to zero far below zero.

The central gradient tradeoff

For ReLU, $f'(x) = 1$ when $x > 0$ and 0 when $x < 0$. This helps active units pass gradients, but an inactive unit may receive none. Leaky variants preserve a negative-side route when $a > 0$. ELU and SELU have nonzero derivatives at finite negative inputs, yet their negative tails still saturate. GELU smooths the transition rather than applying a hard sign gate.

Activation curves

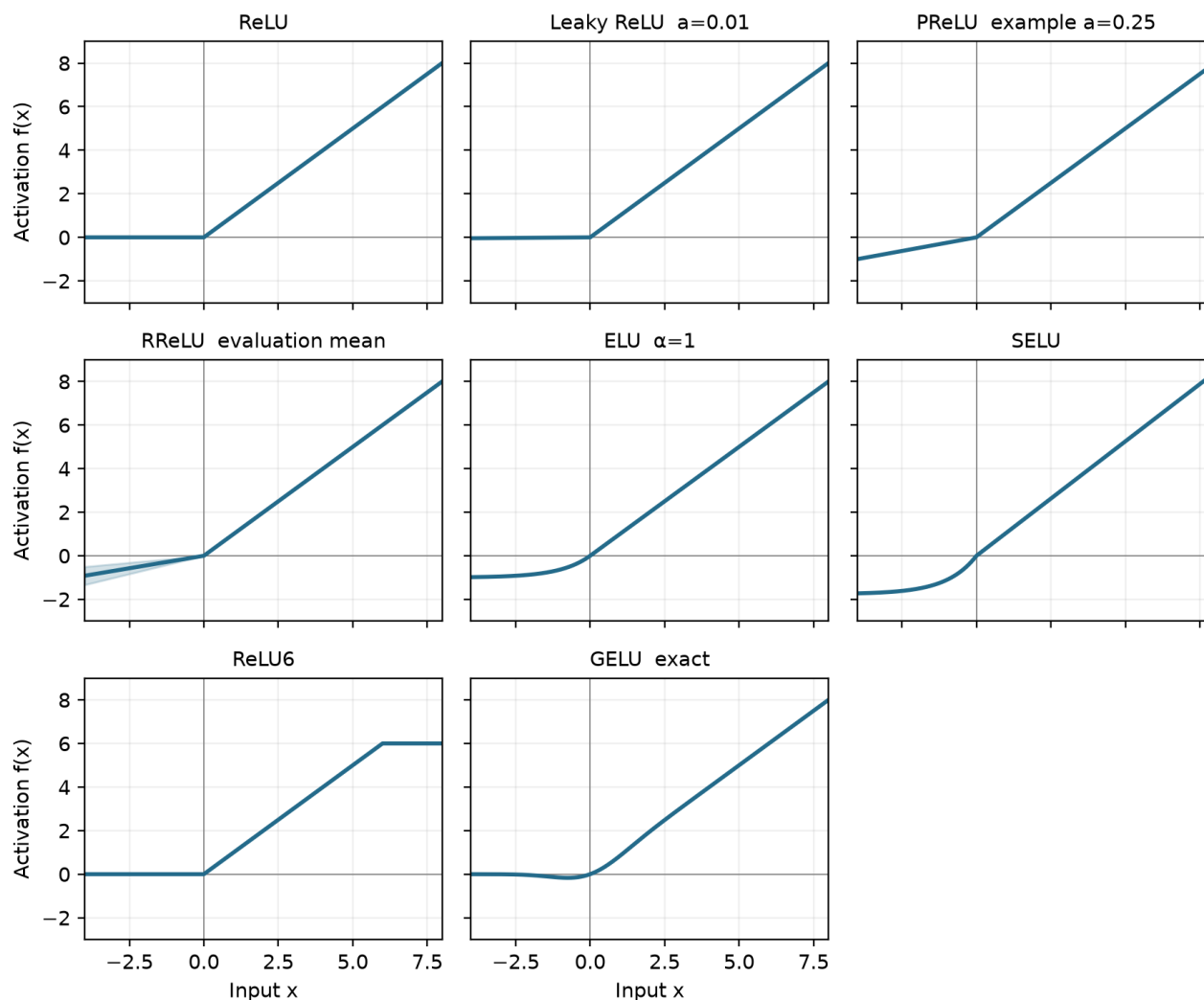


Figure 1. Exact activation curves with shared scales. PReLU uses an illustrative learned slope of 0.25, not a training result. RReLU shows its evaluation curve with $l=1/8$ and $u=1/3$; the shaded band is the possible negative branch during training. ELU uses $\alpha=1$. GELU uses the exact normal CDF.

Derivatives and smoothness

$$\text{ELU}'(x) = \alpha e^x \quad (x < 0), \quad \text{GELU}'(x) = \Phi(x) + x\phi(x)$$

ϕ denotes the standard-normal density. GELU is smooth and non-monotonic; its derivative can be negative for some negative inputs and approaches zero in the far negative tail. It therefore avoids ReLU's entire zero-gradient half-line without guaranteeing a useful gradient everywhere. ELU has matching derivatives at zero only when $\alpha=1$; SELU has different one-sided slopes at zero.

ReLU6 has derivative 1 within $(0,6)$, and zero outside it. Thus clipping adds positive-side saturation. PReLU slopes are trainable parameters and are not inherently constrained to stay positive; if the learned slope approaches zero, the usual negative-side benefit can weaken.

Advantages limitations and typical uses

Function	Gradient and cost	Benefits and limitations	Typical application
ReLU	Positive gradient 1; cheapest	Fast and sparse; dead-unit risk; unbounded positive outputs	General MLP and CNN baseline
Leaky ReLU	Negative gradient a ; low cost	Reduces hard inactivity; fixed slope needs selection	CNNs and adversarial models
PReLU	Learned negative slope; low cost plus parameters	Adaptive response; added fitting capacity and slope drift	Deep image classifiers
RReLU	Random slope; sampling cost in training	Noise can regularize; train/eval behavior differs	Small-data CNN experiments
ELU	$\alpha \exp(x)$ below 0; exponential cost	Negative outputs reduce mean shift; negative-tail saturation	Deep CNN or MLP alternatives
SELU	Scaled ELU derivative; exponential cost	Conditional self-normalization; requires compatible initialization and dropout	Feedforward self-normalizing networks
ReLU6	Zero outside (0,6); very low cost	Bounded range can aid low-precision design; clipping discards magnitude	Mobile or quantization-oriented networks
GELU	Smooth $\Phi(x) + x\phi(x)$; higher cost	Smooth value-dependent gating; negative-tail gradients small	Transformer feedforward layers and NLP

Practical examples and training performance

At $x=-2$: ReLU=0, Leaky ReLU($a=0.01$)=-0.02, PReLU($a=0.25$)=-0.50, ELU($\alpha=1$) $\approx$ -0.865, and GELU $\approx$ -0.0455. At $x=8$, ReLU outputs 8 while ReLU6 outputs 6. These values illustrate leakage, saturation and clipping; they do not rank training accuracy.

Use He initialization for ReLU and adjust its gain for leaky slopes. SELU commonly uses LeCun normal, standardized inputs and AlphaDropout. GELU is conceptually related to rectifiers but is not a piecewise-linear ReLU. Compare variants using the same data split, optimizer and tuning budget, then repeat seeds; architecture and data determine the winner. In the LSTM project, internal gates use sigmoid/tanh, while the six independent outputs use sigmoid probabilities.